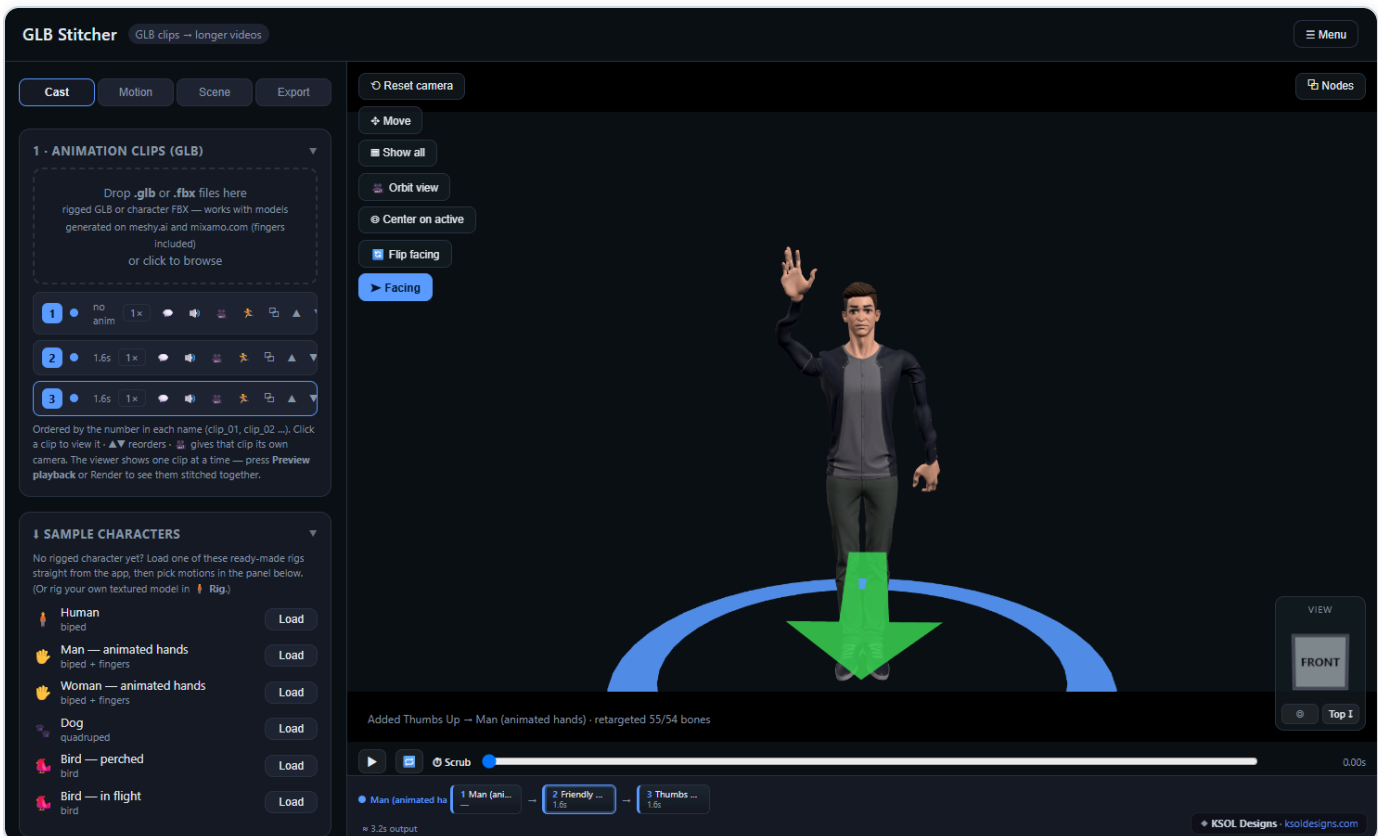


GLB Studio

User Manual · from a picture to an animated character, videos, boards and games, in your browser

KSOL DESIGNS · VERSION OF OCTOBER 1, 2026



Free to use at ksoldesigns.com/glb-studio. Everything runs in your web browser; your models stay on your computer.

What's inside

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▶ WATCH IT



Prefer to watch first? Scan the code or click: [The complete how-to, all six steps \(10 min\)](https://youtu.be/TVR_tN5dO9Y)
youtu.be/TVR_tN5dO9Y

[How-To #1: Generate](#)
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THE IDEA IN ONE LINE

Each tab hands its work to the next through a shared **Studio Library**: **Generate** a model (or bring your own) → **Rig** it → **Animate** motions → **Stitch** them into a video, or put the character on a board in **Tabletop** and play, or plan a scene from above in **Maestro**. There are no files to shuffle between steps.

QUICK START (5 MINUTES)

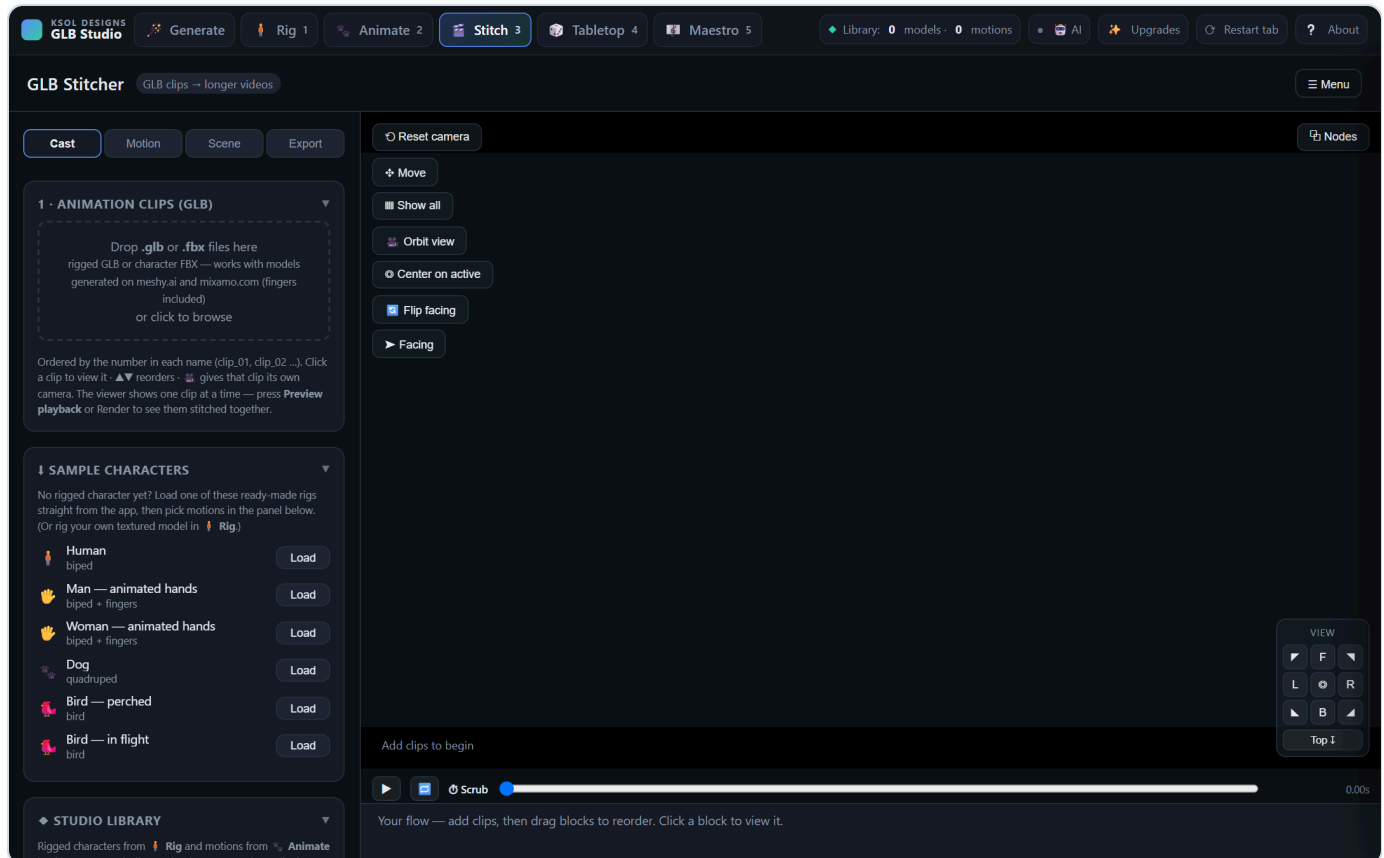
Open 🎬 **Stitch** → under **Sample characters** press **Load** next to *Man, animated hands* → click two or three motions in the **Motion** tab's library → pick a scene in **Scene** → press ▶ **Render video** in **Export**. You have your first MP4.

1 · Getting started

Opening GLB Studio

Go to ksoldesigns.com/glb-studio (or potassiumsolutions.github.io/glb-studio) in **Chrome** or **Edge** on a computer. Firefox works too. Phones and tablets work for most things, but a computer with a mouse is much more comfortable. There's nothing to install and no account to make.

Prefer to run it from your own computer? Download the project and double-click `run-local.bat`, then open the address it shows.

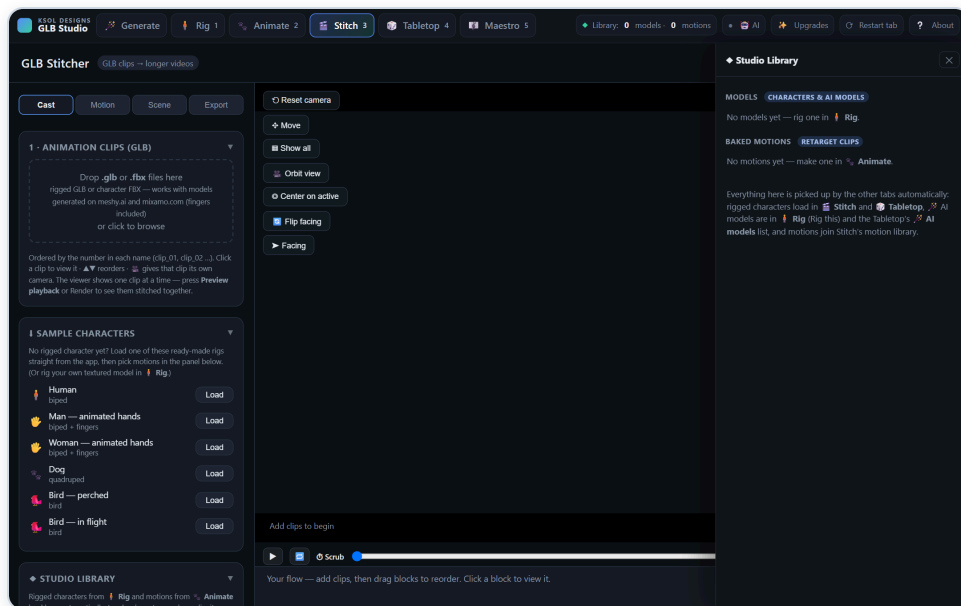


The top bar: the six tabs on the left; on the right ♦ Library, 🤖 AI, ⚡ Upgrades, 🔄 Restart tab and ? About. On smaller screens the right-hand buttons show only their icons; hover over one to see its name.

 Generate	Picture → textured 3D model (an add-on; you pay a few cents per model to fal.ai).
 Rig (1)	Fit a skeleton to a model so it can move.
 Animate (2)	Choose, tune and make motions, and send them to the library.
 Stitch (3)	Put characters, motions, scenes and props together and render an MP4.
 Tabletop (4)	Boards, maps, cards, boxes and minis, print-ready, and playable games.
 Maestro (5)	Top-down plays and scenes on a board, with a path for each piece.
 Library	Everything the tabs share: characters, AI models, motions.
 AI	Connect your own AI assistant so it can drive GLB Studio for you (section 8).
 Upgrades	Import add-on keys (<code>.studiokey</code>) (section 9).
 Restart tab	Reloads only the tab you're on, which frees its memory. Your library is kept.
 New version	Appears when GLB Studio has been updated since you opened it. Press it to reload (your library is kept).

The shared library

Anything a tab exports (a rigged character, an AI model, a motion) goes into the **Studio Library** in your browser, and the other tabs pick it up automatically. Click  **Library** to see it and to delete what you don't need. The library lives in **this browser on this computer**. Keep your own copies of anything important (each tab can also download files).



The library drawer.

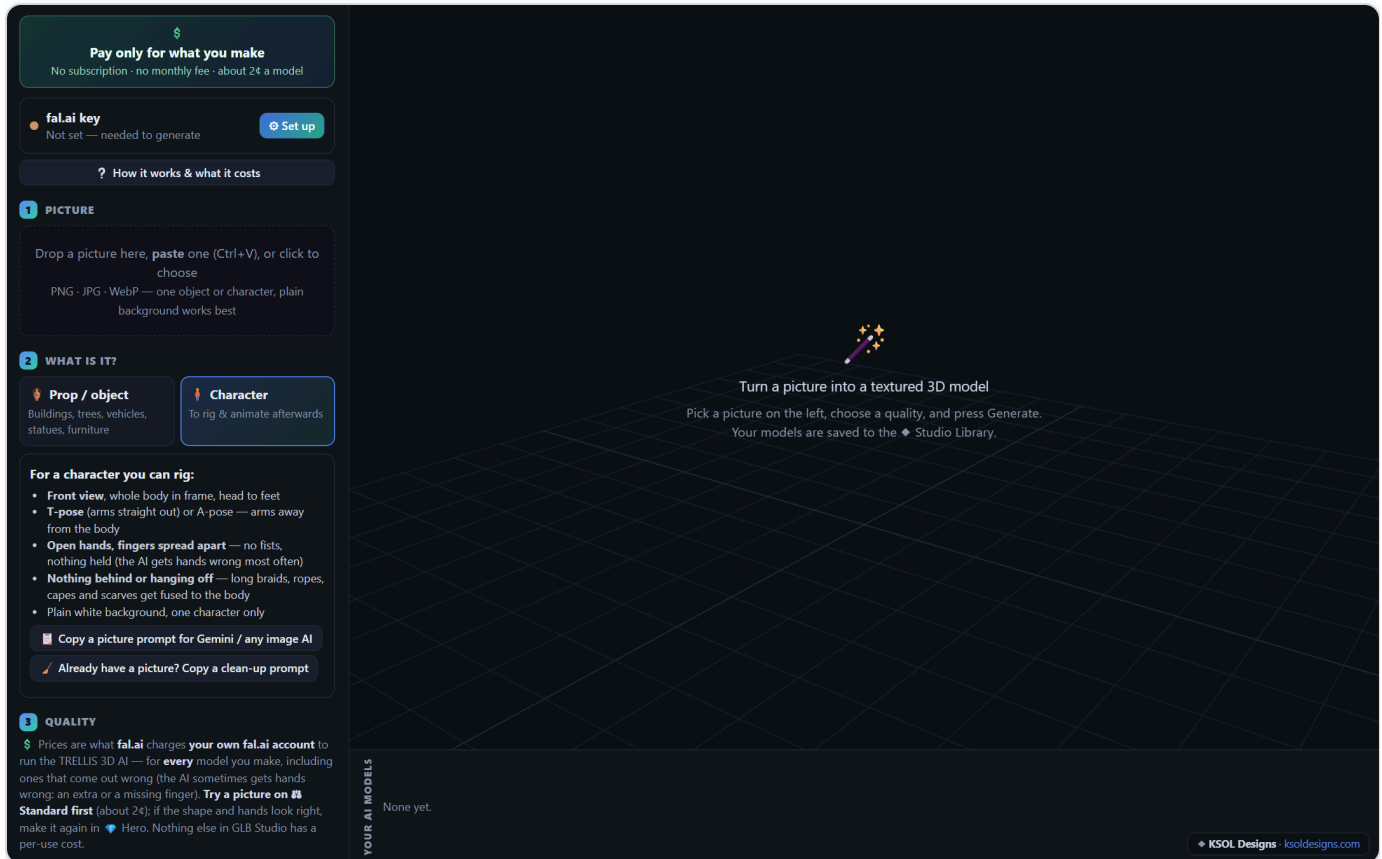
2 · ✨ Generate: turn a picture into a 3D model



▶ WATCH IT

See this chapter in action (about 2 minutes): [How-To #1: Generate](#)
youtu.be/_3DHdxyiMs

An add-on (a one-time purchase; see section 9). Drop in a picture of a character or an object and get a textured 3D model back in a minute or two.



Generate: the picture tips for a character you want to rig, and the two prompts you can copy.

- 1 **Set up once:** press **Set up** and follow the steps: make a free fal.ai account, add a little credit, create a key and paste it. It takes about two minutes.
- 2 **Picture:** drop, paste (Ctrl+V) or choose one, and crop it if needed.
- 3 **What is it?** 🏠 **Prop** (buildings, trees, vehicles, statues) or 🧑 **Character** (to rig and animate).
- 4 **Quality:** ⚡ **Draft** (about 2¢, a quick check), 🛠️ **Standard** (about 2¢, clean paint all over; best for game pieces and big, bulky characters) or 💎 **Hero** (about 25–35¢, sharpest detail for close-ups).
- 5 Press ✨ **Generate**. The model is shrunk to game size and saved to the library. Then press 🧑 **Rig this** or 🎮 **Use in Tabletop**.

Getting a good character

- Use a **front view of the whole body**, head to feet, standing in a **T-pose** (arms straight out) or an A-pose.
- Show **open hands with the fingers spread**. The AI gets hands wrong more often than anything else.
- Leave **nothing hanging behind** the character: long braids, ropes, capes and scarves get fused to the body. Use a plain white background and one character.
- 📋 **Copy a picture prompt** gives you a ready-made prompt for Gemini or any picture AI. ✂️ **Copy a clean-up prompt** fixes a picture you already have.
- **Try a new picture on the Standard first**. If the shape and hands look right, press 💎 **Happy with it? Make it in Hero**. That selects Hero; you still press Generate yourself.
- After each character, GLB Studio **checks the hands for you** (free, on your computer). It warns you if the AI added a finger or tore a hand. Press 👁️ **Look at the hands** for a close-up. If they're wrong, try again; a new try often comes out right.

WHAT IT COSTS

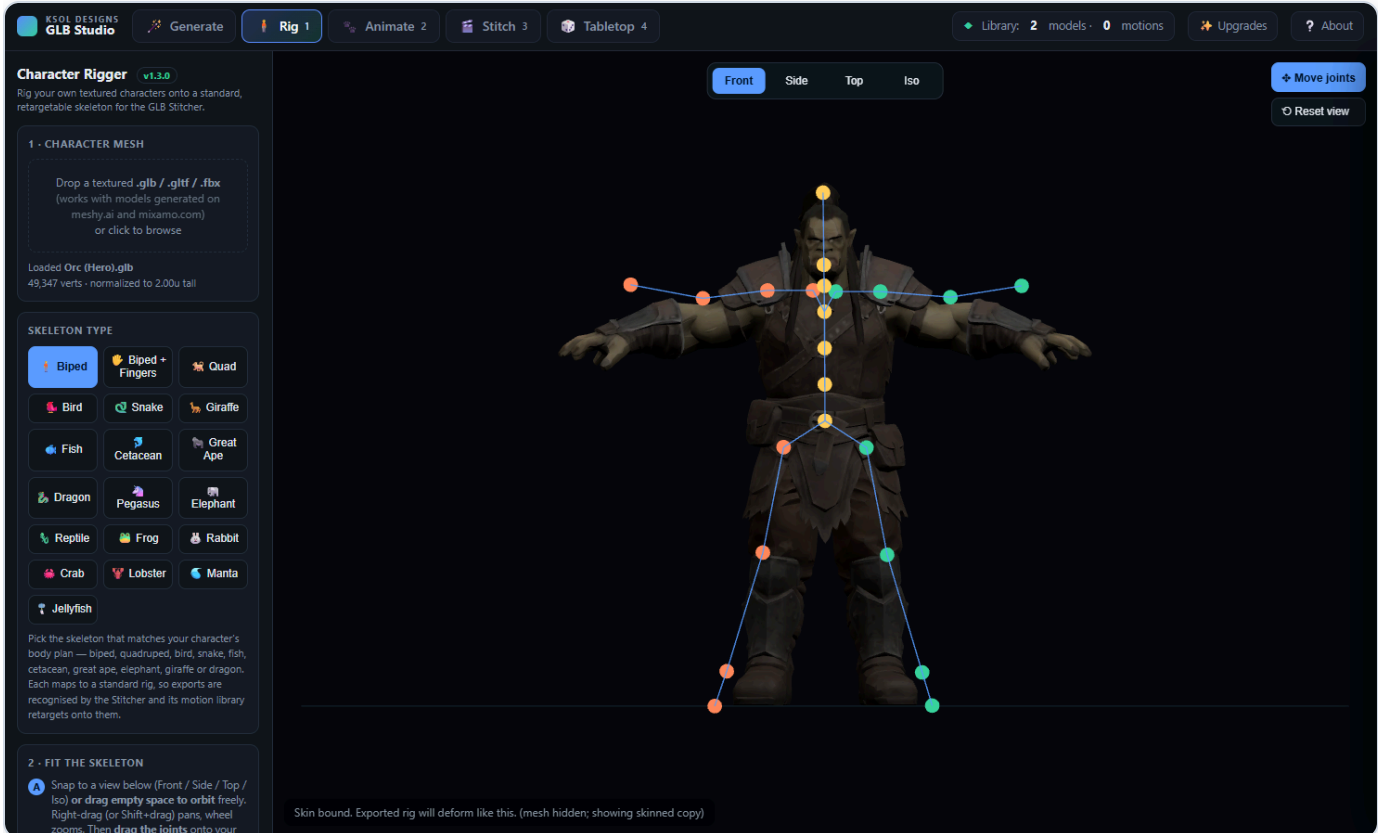
Models are made by the **TRELLIS** AI running on **fal.ai**, and fal.ai charges **your own fal.ai account** for **every** model, including ones that come out wrong. You see the price before you click, and 💰 **Your costs** keeps a running total. KSOL Designs gets none of that money and adds nothing on top. This is the only part of GLB Studio with a per-use cost.

3 • Rig: give a model a skeleton



▶ WATCH IT

See this chapter in action (about 2 minutes): [How-To #2: Rig](#)
youtu.be/nSjJAUKPNS8



Rig: a model generated in GLB Studio, fitted to the biped skeleton automatically and bound, ready to export.

- 1 Drop a textured `.glb`, `.gltf` or `.fbx` (works with models generated on meshy.ai and mixamo.com), or press  **Rig this** in Generate.
- 2 Pick the **skeleton type**. There are 19: biped, biped + fingers, quadruped, bird, snake, fish, whale/dolphin, great ape, dragon, pegasus, elephant, giraffe, reptile, frog, rabbit, crab, lobster, manta ray and jellyfish. For people, **biped + fingers** gives hands that grip, point, clap and wave.
- 3 The skeleton **fits itself** to the model. Drag any joint to fine-tune it;  **Symmetric** copies each change to the other side.
- 4 **Bind & preview weights** shows how the skin follows the bones.
- 5 **Export rigged GLB** downloads the file **and** puts it in the library. Or press  **Open in Stitcher** to go straight on.

TIP

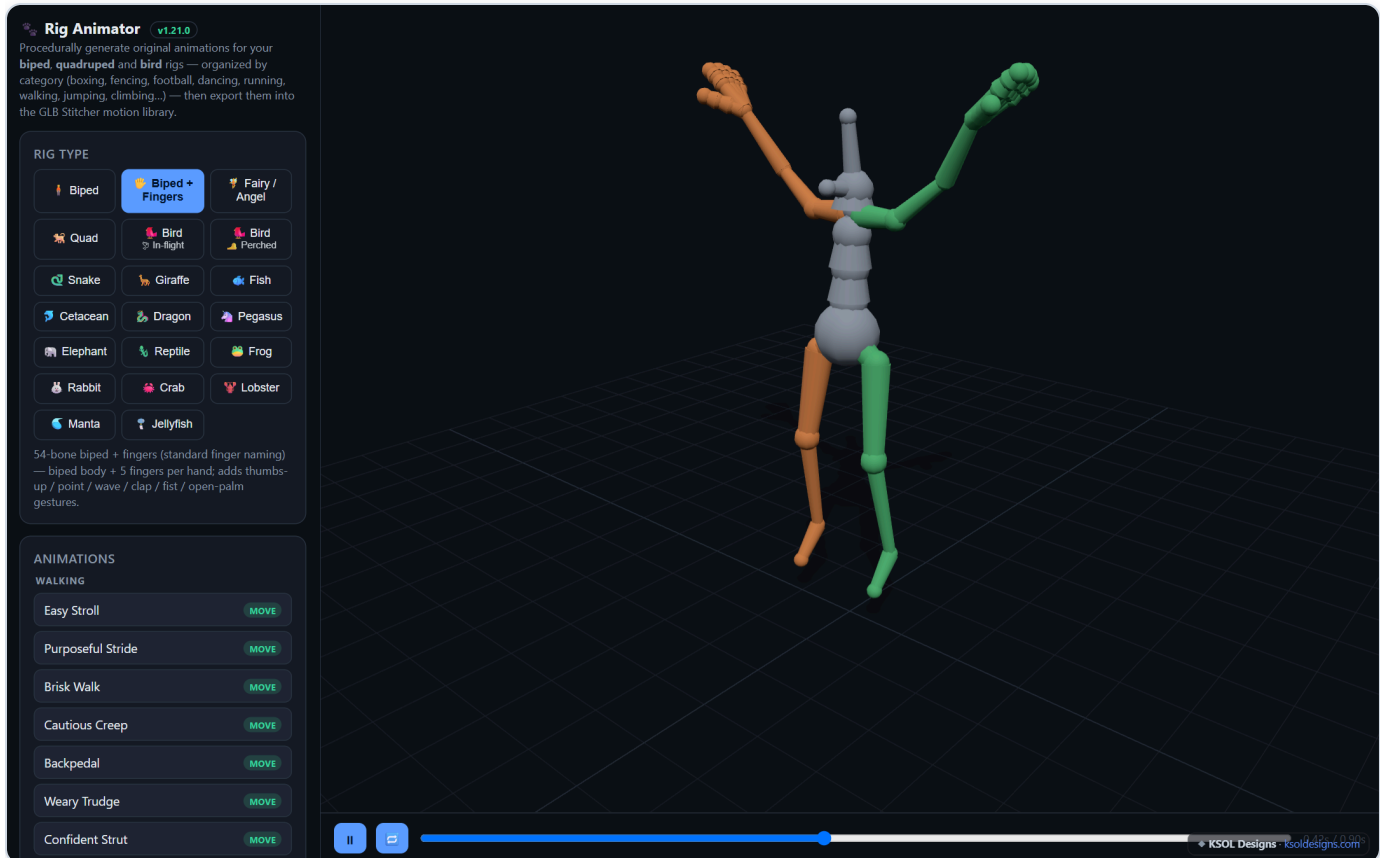
Hands rig best when the fingers are open and apart. If a hand came out as a mitten, choose the plain **biped**: the hand then moves as one piece.

4 • 🐾 Animate: pick, tune and make your own motions



▶ WATCH IT

See this chapter in action (about 2 minutes): [How-To #3: Animate](#)
youtu.be/4VmEKSGQ_O4



Animate: choose the rig type and a motion, tune it, then **+** Add it to the library.

- 1 Choose the **rig type** (the same type you rigged your character with).
- 2 Pick a motion. There are hundreds: walks, runs, jumps, sports, dances, fights, gestures, flying, swimming and more.
- 3 Tune its **speed**, **amplitude** and **loop length**.
- 4 Press **+** **Add current animation** → **Studio Library**. It joins Stitch's motion list for that skeleton.

Three ways to make your own motions

Your motions appear in the list under their own headings (👤 My motions, 📺 From video, 🍷 From a recipe) and are saved on your computer.

🔧 Edit a motion

Start from any motion and press 🔧 **Edit this motion**. Scrub to a moment and **drag the figure**: pull a hand or foot, or turn the head, spine or hips (**Shift**-drag the hips to move the body). Each drag sets a key. Name it and press 💾 **Save as a new motion**. The original is never changed.

🎥 Motion from a video

For people (biped rigs). 📁 **Choose a video** or ● **Record with the camera**: the whole body in the picture, head to feet, with the camera held still, good light and up to 20 seconds. The body, and on a fingered rig the fingers, are tracked **right on your computer** with a free tracker, and your video is never uploaded. Tick **Mirrored video** for a selfie camera. One camera has to guess depth, so look at the result from the side and touch it up with 🔧 if needed.

➡ MAKE A NEW MOTION

Start from the selected motion, then at any moment on the timeline **drag the figure**: pull a hand or foot, turn the head, spine or hips (**Shift**-drag the hips to move the body). Each drag sets a key; the motion blends between keys. The original is never changed.

🔧 Edit this motion

🎥 MOTION FROM A VIDEO

Film someone moving — the **whole body in the picture** (head to feet), the camera held **still**, good light, up to 20 seconds. GLB Studio tracks the body — and the fingers, on a fingered rig — **right here on your computer** (free; the video is never uploaded) and turns it into a new motion in 🎥 **From video**.

📁 Choose a video

● Record with the camera

☐ Mirrored video (a selfie camera) — swap left and right

Name

My video motion



Motion from a video: each video frame (left of each pair) next to the pose it became (right).

Motion from a description

For people. Press  **Copy the instructions for your AI**, paste them into any AI chat (ChatGPT, Claude, Gemini...) and finish the last line with what you want, for example *"a proud victory jump with both fists up"*. The AI answers with a **motion recipe**, a few key poses written as text. Paste it into the box and press  **Make the motion**. Not quite right? Ask the AI to change it and paste the new recipe. To have your AI do all of this by itself, see section 8.

MOTION FROM A DESCRIPTION

Describe a motion to **any AI chat** (ChatGPT, Claude, Gemini...): press  **Copy the instructions**, paste them into the chat, add what you want (*"a proud victory jump with both fists up"*), then paste its answer (a **motion recipe**) here. Or connect your AI to GLB Studio ( AI, top right) and it does all of this for you.

 **Copy the instructions for your AI**

Paste the motion recipe here – it starts with `{"name": ...`

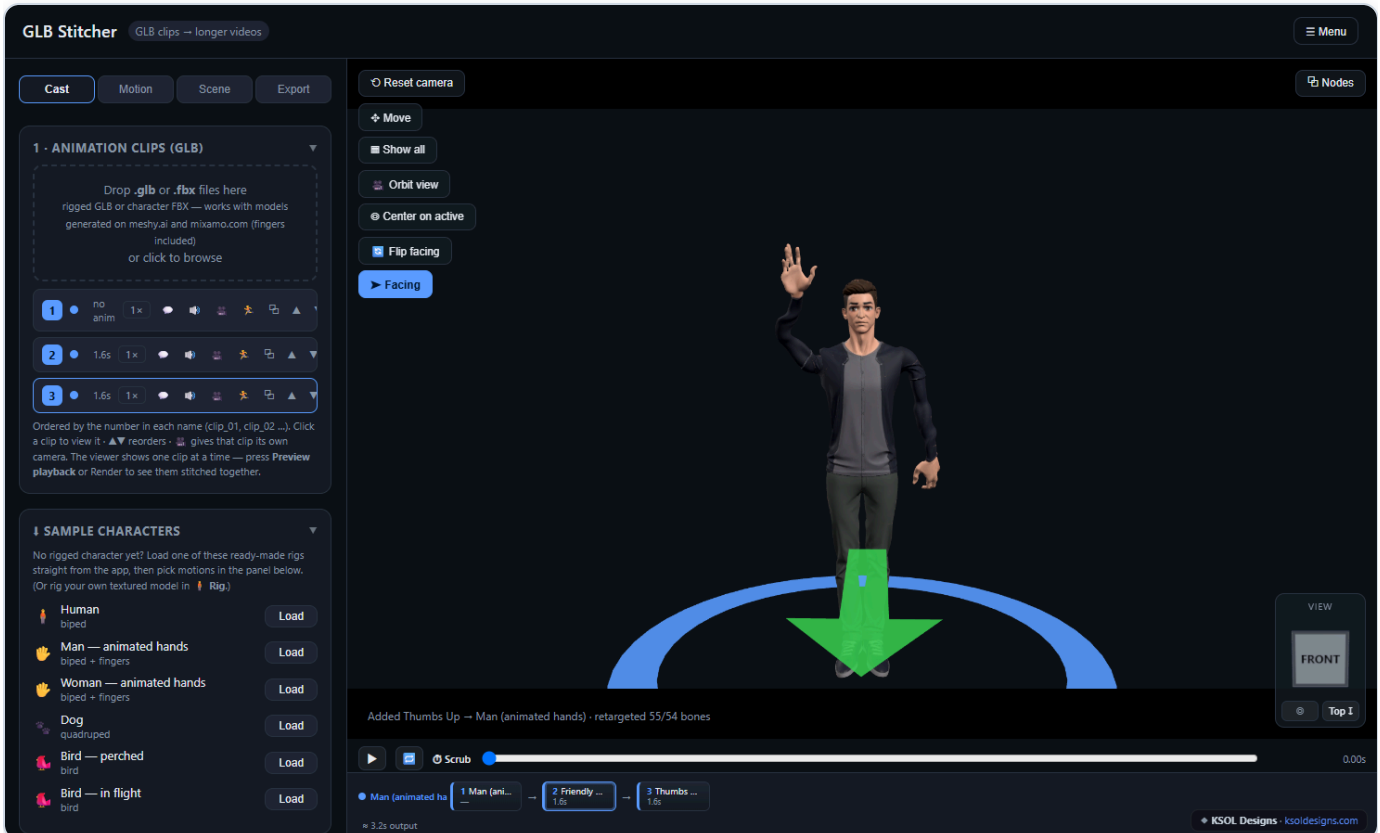
 **Make the motion**

5 • Stitch: make the video



▶ WATCH IT

See this chapter in action (about 2 minutes): [How-To #4: Stitch](https://youtu.be/SdMR69cmNpE)
youtu.be/SdMR69cmNpE



Stitch: a sample character with three motions, in a scene, ready to render.

- 1 **Cast:** load a rigged character from the ♦ **Studio Library** card, drop in a `.glb`, or press **Load** on a free sample.
- 2 **Motion:** the library shows only the motions that fit that character. Click one to add it as a clip; it's **retargeted** onto your character automatically. Reorder the clips, change their speed and loops, and set cross-fades.
- 3 **Scene:** pick a set (fields, forests, cities, rooms...), props for the scene, and a prop for the hand (sword, staff, spear, bat...). On characters with fingers, the fingers close around the handle.
- 4 **Camera:** use the **view cube** in the bottom-right corner of the 3D view: click a face, edge or corner to look from that side, or drag the cube to circle the camera. © resets the view and **Top** looks straight down. (A plain drag on the 3D view moves the character; hold **Alt** or use **Orbit view** to orbit with the mouse instead.) Let the camera follow the action.

- 5 **Export:** choose the size (for example 1080×1920 for phones), add a title card if you like, and press ►
Render video for an MP4.

MORE THAN ONE CHARACTER

Add more characters in Cast, and each gets its own lane of clips. They play together on the same stage.



Send to Maestro takes the clips and sound over to Maestro.

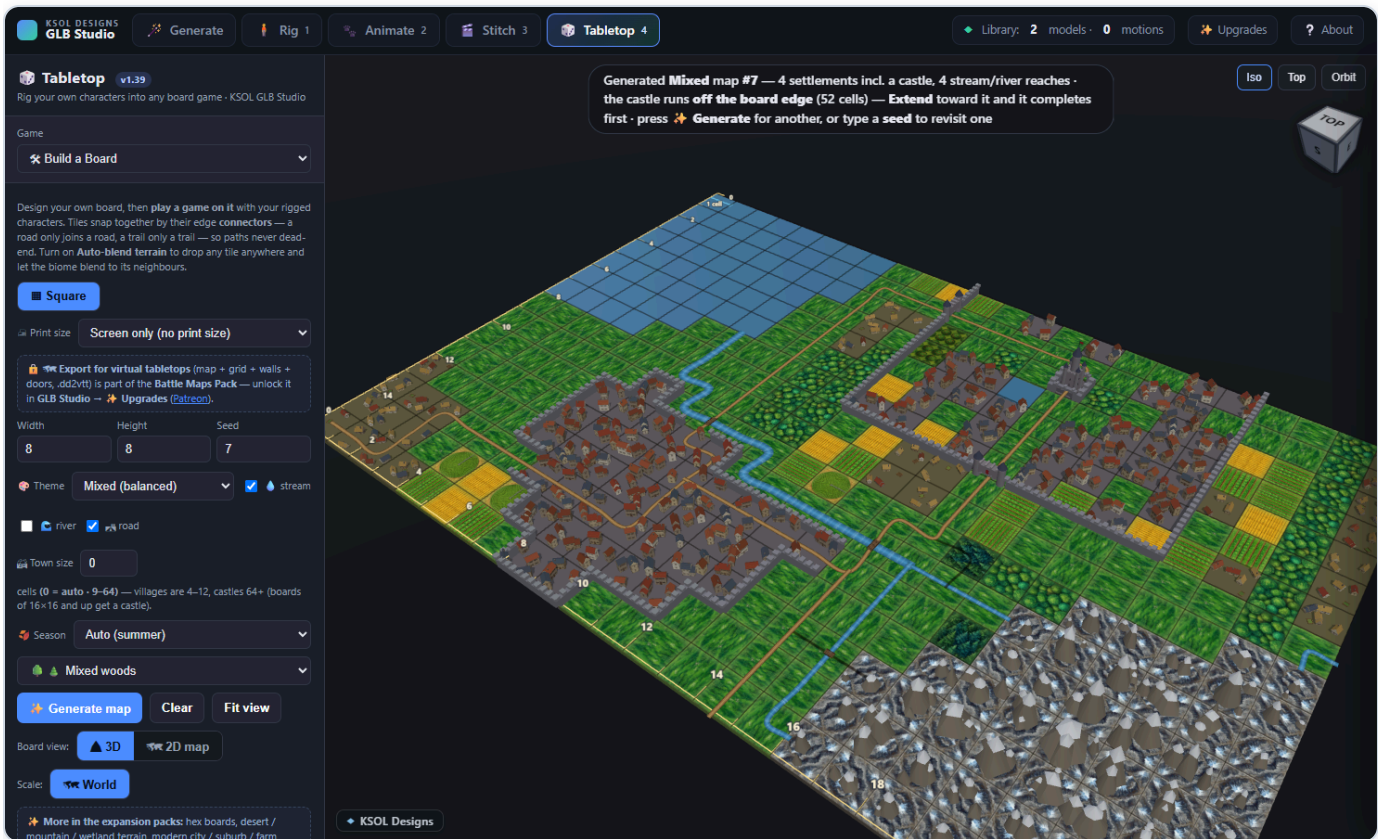
6 • 🎲 Tabletop: make the content for your game



▶ WATCH IT

See this chapter in action (about 2 minutes): [How-To #5: Tabletop](#)
youtu.be/cbMt-WRC2yM

Maps and boards, cards, boxes and minis: print-ready for The Game Crafter or a print shop, ready for virtual tabletops, or built into your own app. Play-test it right there with your characters as the pieces. Use the **Game** list at the top left:



Tabletop → Build a Board: a generated map with a walled castle, villages, a stream, roads, farms and mountains.

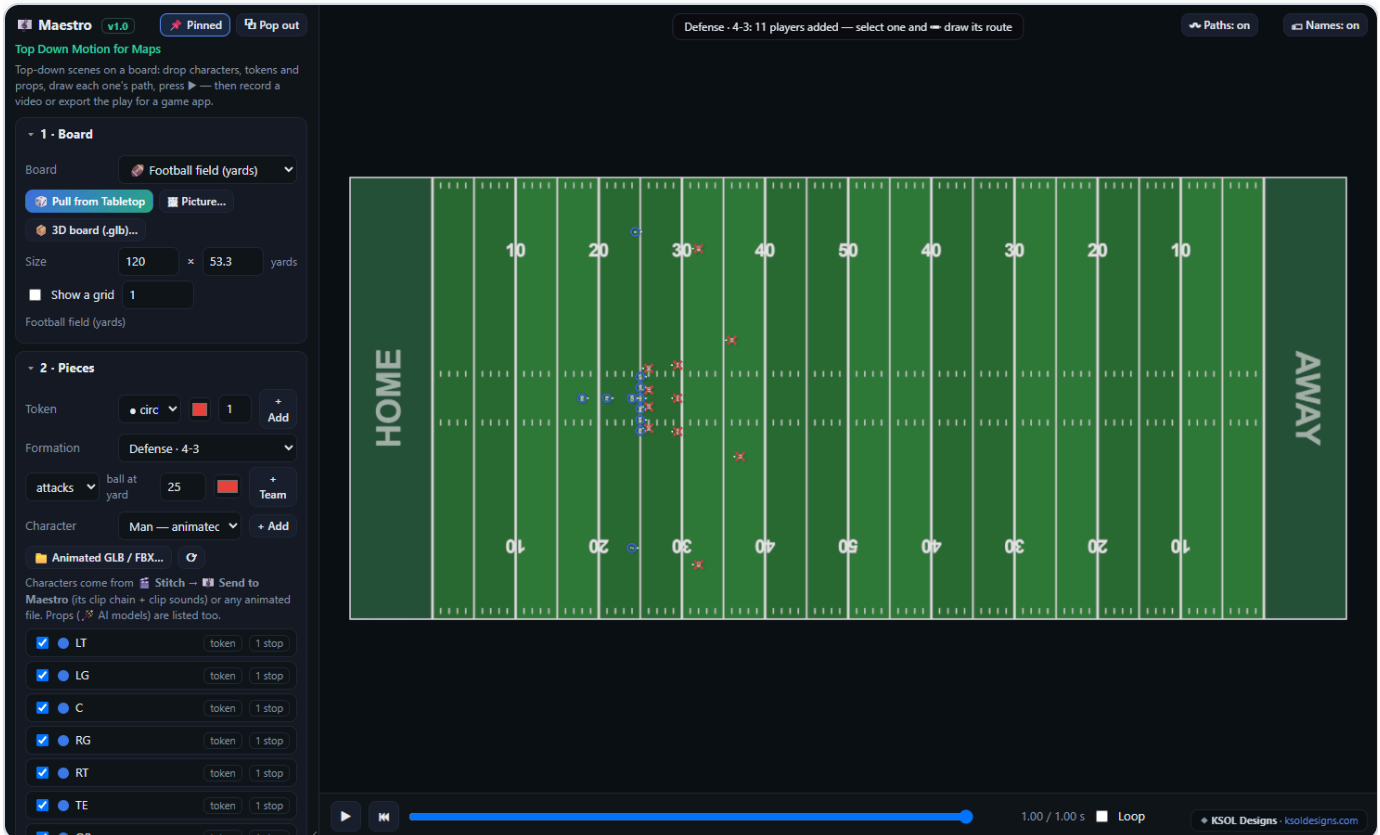
- **Chess, Checkers and Chinese Checkers** (free), plus race games like the Royal Game of Ur (Race Games pack), played with your own rigged characters. Set up the teams and play the computer or a friend.
- **🔧 Build a Board:** generate a map (grassland, forest, towns, castles, rivers, roads and more) with one click, **Extend** it in any direction, paint tiles and place props, including your own and 🧙 AI models. Choose a print size to export print-ready art (The Game Crafter's boards and mats, or one-piece posters up to 48×36 inches).
- **🃏 Cards** (a deck designer) and **📦 Box designer** (Game Crafter packaging templates).
- **📱 Export playable app:** turns your game into a small app you can share and run on its own.

7 • 🎵 Maestro: top-down motion for maps



▶ WATCH IT

See this chapter in action (about 2 minutes): [How-To #6: Maestro](#)
youtu.be/3Co917sUURk



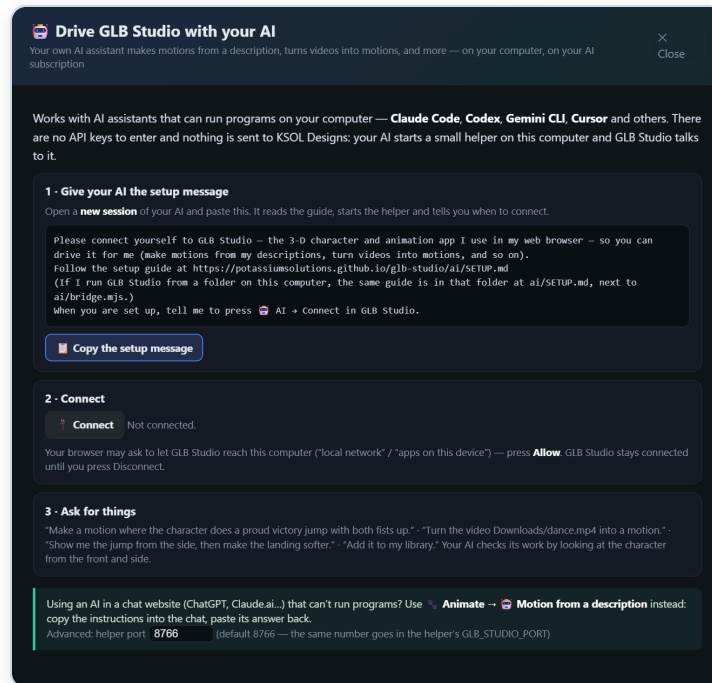
Maestro: a football field with an offense and a 4-3 defense, each added in two clicks. Select a player and draw its route.

- 1 **Board:** pick one (a sports field, a grid, a picture) or press 🎲 **Pull from Tabletop** to use the board you built.
- 2 **Pieces:** add tokens, **formations** (+ **Team** adds a whole side) or characters: the ones you sent from Stitch, or any animated GLB/FBX.
- 3 **Paths:** select a piece and draw its route as a chain of points. Each point can have its own timing and motion.
- 4 Press ▶ to watch the play. **Record** a video, or export the play as path data (JSON/CSV) or an animated GLB for a game or app.

8 • Drive GLB Studio with your own AI

If you use an AI assistant that can run programs on your computer (**Claude Code**, **Codex**, **Gemini CLI**, **Cursor** and others), it can drive GLB Studio for you. It makes motions from your descriptions and checks them from the front and side, turns your videos into motions, and adds them to your library.

There are **no API keys** and nothing is sent to KSOL Designs. Your AI downloads one small helper file, runs it on your computer on your own AI subscription, and GLB Studio talks to it.



The  AI panel: copy the setup message, then Connect.

- 1 Click  **AI** (top right) and press  **Copy the setup message**.
- 2 Open a **new session** of your AI and paste it. The AI reads the setup guide, checks you have Node.js (it will ask you to install it from nodejs.org if not), downloads the helper and starts it. This takes a couple of minutes.
- 3 When it says it's ready, press  **Connect** in the panel. The first time, your browser asks to let GLB Studio reach "**apps on this device**" (the local network): press **Allow**. The dot on the  AI button turns **green**.
- 4 Ask for things in plain words.

Things to ask

- "Make a motion where my character waves hello with the right hand while bouncing at the knees."
- "Show me the jump from the side, and make the landing softer."
- "Turn the video Downloads/dance.mp4 into a motion." (It must be a video file on your computer.)
- "Add it to my library." Then use it on any character in  **Stitch**.



What your AI sees when it checks its work: the front...



...and the side, at several moments of the motion.

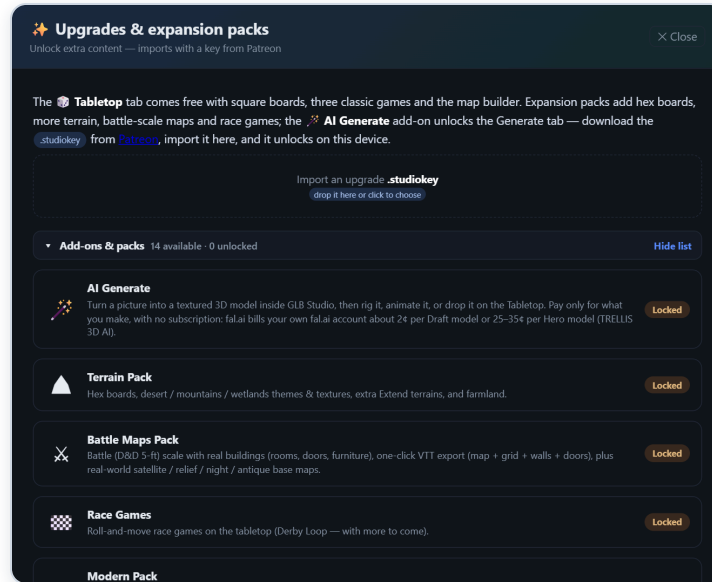
IF IT WON'T CONNECT

The panel says "Waiting for your AI...": the helper isn't running yet, so ask your AI to start it. **You pressed Block by mistake:** click the  or settings icon left of the web address, allow **apps on this device** (local network), and reload. **Safari** sometimes refuses this connection; use Chrome, Edge or Firefox. **An AI in a chat website** (ChatGPT, Claude.ai) can't run programs, so use  Animate →  Motion from a description instead (section 4).

The helper listens only on your own computer (127.0.0.1) and refuses commands from web pages. It stops when your AI session ends. The setup guide your AI follows is at potassiumsolutions.github.io/glb-studio/ai/SETUP.md.

9 • ✨ Upgrades: add-on keys and expansion packs

The core tools are free. **Add-ons** (✨ AI Generate, and Tabletop add-ons like hex boards and terrain, battle maps, modern, sci-fi and race games) and **expansion packs** (characters, motions, scenes and props) are one-time purchases on patreon.com/KsolDesigns.



✨ Upgrades: drop a .studiokey here.

Add-on keys (.studiokey)

- 1 Click ✨ **Upgrades** (top bar).
- 2 Drop the `.studiokey` file onto **Import an upgrade** (or click it and choose the file).
- 3 You'll see ✅ **Unlocked**. The Tabletop map packs (Sci-Fi, Pharaohs, Horror and others) also need the **Battle Maps** key.

Expansion packs (.zip)

- 1 Unzip the download. Inside is one `.glbpack.json` file (and the license).
- 2 In 📁 **Stitch**, in the **Motion** library panel, drop the `.glbpack.json` onto 📁 **Expansion packs** (or click it and choose the file).
- 3 Its characters, motions, props and scenes appear. The big character packs take a minute to import.

UPDATES TO PACKS

When a pack is improved, download it again from its Patreon post and import it the same way. The new version replaces the old one automatically. Keys and packs are saved in the browser you imported them into; keep the files so you can import them again on another computer or after clearing your browser.

10 · Costs, privacy and your work

Costs

- GLB Studio itself is **free**. Add-ons and packs are **one-time** purchases, with no subscription.
- The only per-use cost is  **Generate**: fal.ai charges your own fal.ai account a few cents per model (see section 2).
- Driving GLB Studio with your own AI uses **your** AI subscription; GLB Studio adds no cost.

Privacy

- GLB Studio runs in your browser, and your models, motions and library stay on your computer.
-  **Generate** sends the picture you choose to fal.ai to make the model. Your fal.ai key is sent only to fal.ai.
-  **Motion from a video** runs on your computer. The free tracker is downloaded once, and your video is never uploaded.
-  **AI** only talks to a helper on your own computer.
- Nothing goes to KSOL Designs.

What you may do with your work

Everything you make with GLB Studio is **yours to share and sell**: videos, animations, images, boards, maps, cards, boxes, games, apps, prints and 3D prints. You don't pay any royalty or extra license. The one thing you can't do is share or resell the pack and key files themselves. Credit ("Made with GLB Studio · KSOL Designs") is appreciated but never required. The full license comes in every pack (`LICENSE.txt`).

11 · Troubleshooting and tips

What you see	What to do
A  New version button	GLB Studio was updated. Press it to reload; your library is kept.
A tab is slow or stuck	Press  Restart tab . It reloads only that tab.
My character isn't in Stitch's motion list, or its motions are missing	Motions only show for the skeleton they fit. A character rigged with biped + fingers shows the finger motions; a plain biped shows the plain ones.
Generate: the hands are wrong	Generate again (a new try often comes out right), or clean up the picture with the  prompt: open hands, fingers apart. Test new pictures on  Standard before  Hero.
Generate: patches with no paint on Hero	These happen on big, bulky characters. They're filled automatically; for those characters use  Standard.
A video motion looks flat or twisted from the side	One camera guesses depth. Film again with the whole body in view and the camera still, facing the person, or fix the pose with  Edit.
My pack or key disappeared	They're stored in the browser. After clearing browser data, or on another computer, import the files again.
 AI won't connect	See the box on the AI page (section 8).

Tips

- **Save your work** as files too (Rig's ↓ Export, Stitch's MP4, Tabletop's exports). The library lives only in this browser.
- For phone videos, render at **1080×1920**. For YouTube, use 1920×1080.
- Check a new motion from the **front and the side** before you use it. The side view shows the things the front hides.

QUESTIONS, IDEAS, PROBLEMS

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GLB Studio works with models generated on meshy.ai and mixamo.com and with fal.ai. It is not affiliated with or endorsed by those services, Microsoft, Google, Anthropic, OpenAI or the other companies named here. All trademarks belong to their owners.